

COS80022 – Software Quality and Testing

Assignment 2 (D&HD)

Student Name: Ajula Peedikaparambil Raju
Student ID: 105286562
Student email: 105286562@student.swin.edu.au
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A. Review of Requirements from a Tester's Perspective

A.1 Boundary Testing

Classic Saver Deposit

Input (Deposit Amount)	Expected System Behaviour	Reasoning
\$499.99	Rejected -- Below minimum allowed	Below Classic Saver minimum (\$500) stated in Requirement 9.
\$500.00	Accepted -- Minimum valid deposit	Meets Classic Saver table requirement but conflicts with global rule (\$1000 minimum).
\$999,999.99	Accepted -- Maximum valid deposit	Within upper limit defined in the requirement.
\$1,000,000.00	Rejected -- Exceeds maximum	Exceeds stated maximum of \$999,999.99.

Premium Saver Deposit

Input (Deposit Amount)	Expected System Behaviour	Reasoning
\$2,499.99	Rejected -- Below minimum allowed	Below Premium Saver minimum of \$2,500.
\$2,500.00	Accepted -- Minimum valid deposit	Meets requirement-defined minimum for Premium Saver.
\$999,999.99	Accepted -- Maximum valid deposit	Within upper limit (\$999,999.99).
\$1,000,000.00	Rejected -- Exceeds maximum	Beyond defined range.

Supplementary Account Holders

Input (Number of Holders)	Expected System Behaviour	Reasoning
-1	Rejected -- Invalid (below range)	Negative values are logically invalid for count-based input.
0	Accepted -- Minimum valid	Allowed per requirement: between 0 and 3 supplementary holders.
3	Accepted -- Maximum valid	Upper limit specified in the requirement.
4	Rejected -- Exceeds maximum	Violates 'maximum of three' rule.

A.2 Review Against Boundary Testing Questions

1. Can boundary values be fully defined?

No. Classic Saver deposit boundaries cannot be fully defined because of the conflict between the \$500.00 minimum (table) and the \$1,000 minimum (text). This ambiguity prevents clear interpretation. All other boundaries (Premium Saver, supplementary holders) are well defined.

2. Are outputs defined for all boundaries?

No. Requirements specify acceptable ranges but do not define expected outputs or error messages for invalid entries. For instance, the rejection message or system response for an invalid deposit or holder count is unspecified.

3. Do any boundaries have more than one output defined?

Yes. Classic Saver deposits between \$500 and \$999.99 could either be accepted (based on table) or rejected (based on textual rule). This results in multiple possible outputs and represents a requirement inconsistency.

B. Decision Table Testing

B.1 Decision Table for SmartGym System

Conditions / Actions	Rule 1	Rule 2	Rule 3	Rule 4
Member has Gold Membership	Yes	Yes	No	No
More than 15 sessions booked	Yes	No	Yes	No
Actions / Expected Discounts				
Apply combined 12% discount	X			
Apply 7% Gold discount only		X		
Apply 5% session discount only			X	
Apply no discount				X

B.2 Derived Test Cases

Test Case ID	Membership Type	Sessions Booked	Expected Discount	Expected Result
TC1 (Rule 1)	Gold	20 (>15)	12%	Combined discount applied.
TC2 (Rule 2)	Gold	10 (≤ 15)	7%	Only Gold discount applied.
TC3 (Rule 3)	Basic	18 (>15)	5%	Session discount applied.
TC4 (Rule 4)	Basic	12 (≤ 15)	0%	No discount.

C. State Transition Testing

C.1 Valid Single State Transitions

Test ID	Start State	Event	Action	End State
ST1	Not Started	Login	Display Dashboard	Dashboard
ST2	Dashboard	Start Registration	Display Courses	Course Selection
ST3	Course Selection	Member Info	Display Member Info	Member Info
ST4	Member Info	Proceed to Checkout	Display Fees	Fee Payment
ST5	Fee Payment	Payment	Display Payment Details	Payment Confirmation
ST6	Course Selection	Cancel	Display Dashboard	Dashboard
ST7	Member Info	Back	Display Course Selection	Course Selection
ST8	Fee Payment	Back	Display Member Info	Member Info
ST9	Payment Confirmation	Close Payment Details	Display Course Selection	Course Selection

C.2 Invalid Transitions

Start State \ Event	Login	Start Registration	Member Info	Proceed to Checkout	Payment	Cancel
Not Started	Valid	Invalid	Invalid	Invalid	Invalid	Invalid
Dashboard	Invalid	Valid	Invalid	Invalid	Invalid	Invalid
Course Selection	Invalid	Invalid	Valid	Invalid	Invalid	Valid
Member Info	Invalid	Invalid	Invalid	Valid	Invalid	Invalid
Fee Payment	Invalid	Invalid	Invalid	Invalid	Valid	Invalid
Payment Confirmation	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid

Invalid Transition Examples:

- Attempting to Start Registration from Not Started state should be rejected (user must log in first).
- Attempting to Login again when already in Dashboard state should be invalid.
- Attempting to Proceed to Checkout directly from Dashboard without going through Course Selection and Member Info should be blocked.
- Attempting to make Payment when in Course Selection state should be rejected (must proceed through Member Info and Fee Payment first).
- Attempting to use Back event from Dashboard or Not Started states should be invalid.